

Yuhang Zhang

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EDUCATION

Texas A&M University

College Station, TX

Ph.D. Candidate, Mechanical Engineering; GPA: 4.0/4.0

Jan 2023 – Feb 2027 (Expected)

- Dissertation: *Dynamic Modeling and Optimal Control of District Heating and Cooling Systems (working title)*
- Advisor: Prof. Zheng O'Neill

Tongji University

Shanghai, China

Master of Engineering in Power Engineering; GPA: 4.9/5.0

Sep 2019 – Mar 2022

- Thesis: *Study on Leakage Diagnosis Method for Air Conditioning Water System Networks*

Tongji University

Shanghai, China

Bachelor of Science in Building Environment and Energy Engineering; GPA: 4.6/5.0

Sep 2015 – Jul 2019

RESEARCH INTERESTS

- District and Building Energy Systems — 5GDHC, HVAC&R, heat pumps, and thermal energy storage
- Intelligent Controls and Optimization — model predictive control and data-driven methods for thermal systems
- Building Decarbonization and Electrification — renewable energy, geothermal systems, and waste heat integration

RESEARCH EXPERIENCE

NSF PIRE: Building Decarbonization via AI-empowered District Heat Pump Systems

National Science Foundation (NSF)

Jan 2023 – Present

- Developed a high-fidelity virtual testbed of district heat pump systems integrating physics-based Modelica thermal models with AI-driven predictive control frameworks.
- Designed Model Predictive Control (MPC)-based optimal control frameworks for intelligent district energy system operation.
- Applied the framework to assess renewable and waste heat integration (e.g., data center heat recovery), demonstrating improved efficiency and carbon reduction potential.

DoD ESTCP: Building Energy Efficiency through Thermal Microgrids – Fort Hood Phase I Feasibility Study

Department of Defense (DoD) ESTCP

Dec 2023 – Present

- Developed and validated Modelica digital twins and machine-learning load forecasting models for 30+ buildings.
- Implemented model predictive control optimization for plant operation and temperature reset strategies.
- Produced scenario-specific simulations projecting nearly 68% lower annual district thermal-system energy input for the best-performing 5GDHC retrofit relative to the existing modeled baseline and performed a 30-year NIST BLCC life-cycle cost analysis.

DOE: Solar-Geothermal District Heating and Cooling (Sol-GDHC) System – Citizen Potawatomi Nation

Department of Energy (DOE)

Dec 2023 – Sep 2024

- Contributed to the design of a 100% electrified geothermal district energy system for an Oklahoma tribal community.
- Developed an integrated simulation framework for system-level performance evaluation and annual operation analysis.
- Developed detailed Modelica component-level models of hybrid photovoltaic thermal (PVT) panels.
- Supported optimization of geothermal borehole design and loop temperature control strategies.
- Produced simulation results for the proposed all-electric single-pipe GDHC retrofit, projecting about 38% lower electricity consumption than the existing heating, cooling, and domestic-hot-water systems and elimination of natural-gas use.

Master's Thesis: Leakage Diagnosis in Air Conditioning Water System Networks

Tongji University

Sep 2020 – Mar 2022

- Modeled thermo-hydraulic dynamics of a complex piping network and validated results against field measurement data.
- Developed a parameter identification approach for pipeline hydraulic resistance using metaheuristic optimization.
- Proposed a 2-stage leakage fault detection and diagnostics method using Adam-optimized neural networks.

PROFESSIONAL EXPERIENCE

PhD Research Intern

Jun 2025 – Aug 2025

Pacific Northwest National Laboratory (PNNL)

Richland, WA

- Validated DOE's TES Sizing Tool by executing 240+ parametric simulations across time-of-use and dynamic pricing tariff scenarios.
- Designed optimized Thermal Energy Storage (TES) control strategies, yielding a simulated operating-cost reduction of up to 85% relative to the original rule-based control in the best-performing scenario.
- Collaborated with DOE's Stor4Build consortium to improve tool robustness for grid-interactive efficient buildings (GEBs).

Graduate Research Assistant

Jan 2023 – Present

Texas A&M University

College Station, TX

- Develop high-fidelity simulations for building and district energy systems using Modelica/Dymola and Python.
- Conduct research on predictive controls and optimization for energy-efficient smart energy systems.
- Contribute to NSF-, DoD-, and DOE-funded research projects on building decarbonization and district energy.

HVAC Engineer

Jul 2022 – Dec 2022

Envision Group

Shanghai, China

- Led development of AHU/chiller control optimization and fault detection & diagnostics (FDD) features for a smart building energy management platform.
- Coordinated with engineering and sales teams to align product design with client needs and deployment requirements.

PUBLICATIONS

Journal Articles

- [1] Liu, M., Zhang, Y., Yang, Z., & O'Neill, Z. (2026). Comparative study of energy flexibility in model predictive control of single- and variable-speed heat pumps with active thermal energy storage. *Energy*, 358, 141387. doi:10.1016/j.energy.2026.141387
- [2] Liu, M., Zhang, Y., Yang, Z., & O'Neill, Z. (2026). A nationwide assessment of energy flexibility from predictive-controlled heat pump and active thermal energy storage system for building electrification. *Building Simulation*, 19(4), 1131–1154. doi:10.1007/s12273-026-1421-4
- [3] Zhang, Y., Liu, M., O'Neill, Z., & Wen, J. (2024). Temperature control strategies for fifth generation district heating and cooling systems: A review and case study. *Applied Energy*, 376, 124156. doi:10.1016/j.apenergy.2024.124156
- [4] Liu, R., Zhang, Y., & Li, Z. (2022). Leakage diagnosis of air conditioning water system networks based on an improved BP neural network algorithm. *Buildings*, 12(5), 610. doi:10.3390/buildings12050610

Conference Proceedings

- [1] Jiang, Z., Song, R., Li, G., Zhang, Y., O'Neill, Z., Wang, X., Goldfeder, J., & Dong, B. (2026). BESTOpt: A modular, physics-informed machine learning based building modeling, control and optimization framework. *SimBuild Conference 2026 (IBPSA-USA)*, 12, 786–797. doi:10.26868/30680611.2026.1367
- [2] Zhang, Y., Liu, M., O'Neill, Z., & Wen, J. (2025). Operating temperature optimization for a 5th-generation district heating and cooling network using model predictive control. *2025 ASHRAE Winter Conference*, Orlando, FL. doi:10.63044/w25zha66
- [3] Li, G., Zhang, Y., Liu, M., O'Neill, Z., Song, L., Wang, G., Cai, J., & Wu, X. (2025). Modeling and integration of photovoltaic thermal (PVT) hybrid solar collectors in a fifth-generation district heating and cooling network (5GDHC). *Proceedings of Building Simulation 2025*. doi:10.26868/25222708.2025.1393
- [4] Zhang, Y., Liu, M., Yang, Z., Calfa, C., & O'Neill, Z. (2024). Development and validation of a water-to-air heat pump model using Modelica. *American Modelica Conference 2024*, 119–126. **Best Student Paper**. doi:10.3384/ECP207119
- [5] Zhang, Y., Liu, M., O'Neill, Z., & Wen, J. (2024). A comparative analysis of energy performance of ultra-low temperature district heating and cooling networks with different substation configurations. *ASHRAE Transactions*, 130, 796–805. doi:10.63044/w24zha94

Under Review

- [1] Guan, H., **Zhang, Y.**, O'Neill, Z., & Dong, B. (2026). Adaptive Fusion: A Scientific Machine Learning Framework for Geothermal Borefield Thermal Modeling. (*Under review at Energy*).
- [2] **Zhang, Y.**, Liu, M., O'Neill, Z., & Wen, J. (2026). A physics-based virtual testbed for fifth-generation district heating and cooling systems: a model predictive control case study of network temperature optimization. (*Under review at Building Simulation*).
- [3] **Zhang, Y.**, Guan, H., O'Neill, Z., & Wen, J. (2026). A unified physics-based framework for system-level fault impact analysis in fifth-generation district heating and cooling systems. (*Under review at Energy & Buildings*).

CONFERENCE PRESENTATIONS (WITHOUT PROCEEDINGS)

- “A Scientific Machine Learning Approach for Geothermal Borefield Thermal Modeling via Residual Learning (co-author)” — *9th International High Performance Buildings Conference (2026 Herrick Conference)*, Jul 12, 2026.
- “Feasibility of Geothermal Borefield Integration in a 5th-Generation District Thermal Energy Network: A Case Study of an Oklahoma Tribal Community” — *Texas A&M Geothermal Research Opportunities Symposium*, Apr 2025. **2nd Place, Poster Competition.**
- “Building Decarbonization with a 5th Generation District Heating and Cooling (5GDHC) System: A Case Study of an Oklahoma Tribal Community” — *Texas A&M Conference on Energy, College Station, TX*, Sep 2024.
- “A Comparative Analysis of Energy Performance of Ultra-low Temperature District Heating and Cooling Networks with Different Substation Configurations” — *BAS2024 – Building Automation Symposium, Syracuse, NY*, May 2024.

ACADEMIC SERVICE

Peer Review

- Reviewer, *Science and Technology for the Built Environment* (2025–Present)
- Reviewer, ASHRAE Conferences (Winter & Annual, 2025–2026)
- Reviewer, AEI International Research & Design Conference (2026)
- Reviewer, *Journal of Engineering for Sustainable Buildings and Cities* (ASME) (2026)

ASHRAE Technical Committees

- Corresponding Member, TC 7.5 – Smart Building Systems (2025–Present)
- Provisional Corresponding Member, TC 6.2 – District Energy (2023–Present)
- Provisional Corresponding Member, TC 6.9 – Thermal Storage (2023–Present)
- Provisional Corresponding Member, TC 6.8 – Geothermal Heat Pump Systems (2023–Present)

Leadership

- ASHRAE Student Branch Officer, Texas A&M University (2023–2024): organized monthly technical seminars and promoted student engagement in ASHRAE activities and conferences.

HONORS & AWARDS

- Texas A&M Mechanical Engineering Continuing Student Fellowship (2026)
- ASHRAE Graduate Student Grant-in-Aid Award (2026–2027)
- J. George H. Thompson Graduate Fellowship (2025)
- Mechanical Engineering Graduate Summer Research Grant (2024–2025, two awards)
- Mechanical Engineering Graduate Student Travel Award (2023–2025, three awards)
- 2nd Place Poster Competition, Texas A&M Geothermal Research Symposium (2025)
- Best Student Paper, American Modelica 2024 Conference (2024)
- Tongji University Scholarships of Excellence (2015–2018, four awards)
- National Encouragement Scholarship (2015–2016)

TECHNICAL METHODS & TOOLS

Programming: Python, Git, MATLAB

Simulation: Modelica/Dymola, FMI/FMU co-simulation, EnergyPlus/OpenStudio

Methods: Model Predictive Control (MPC), Machine learning for thermal systems, Numerical optimization